Encephalopathy Due To Hydroxykynureninuria (KYNU Deficiency): Comprehensive Disease Report

Disease: Encephalopathy Due To Hydroxykynureninuria **MONDO ID:** MONDO:0009372 · **OMIM:** 236800 **Causal gene: KYNU** (kynureninase; HGNC:6469; NCBI Gene 8942; UniProt Q16719; chromosome 2q22.2) **Synonyms:** Xanthurenic aciduria; hydroxykynureninuria; kynureninase deficiency

Summary

The disease catalogued under MONDO:0009372 / OMIM 236800 ("encephalopathy due to hydroxykynureninuria," also called **xanthurenic aciduria** or **hydroxykynureninuria**) is an **autosomal-recessive inborn error of tryptophan metabolism caused by biallelic loss-of-function variants in the gene KYNU**, which encodes the pyridoxal-5′-phosphate (PLP)-dependent enzyme **kynureninase**. Kynureninase catalyzes the hydrolytic cleavage of 3-hydroxy-L-kynurenine to 3-hydroxyanthranilic acid (and, with lower efficiency, of L-kynurenine to anthranilic acid) within the **kynurenine pathway**, the de novo route from tryptophan to nicotinamide adenine dinucleotide (NAD^+). When kynureninase activity is lost, upstream metabolites accumulate and are excreted massively in urine — xanthurenic acid, 3-hydroxykynurenine, and kynurenine — producing the characteristic biochemical signature, while flux to NAD^+ is reduced.

Over nearly six decades the understanding of this entity has transformed. The condition was first described in the 1960s as a benign-appearing biochemical trait ("hydroxykynureninuria"), and the first molecular confirmation came in 2007, when a Somali boy and his brother with massive xanthurenic aciduria were found to be homozygous for a KYNU missense variant. The pivotal conceptual shift came in 2017, when biallelic loss-of-function KYNU variants (alongside variants in HAAO) were shown to cause **Congenital NAD Deficiency Disorder (CNDD)** — a

syndrome of vertebral, cardiac, renal and limb (VCRL) malformations historically overlapping with VACTERL association and Catel-Manzke syndrome. The unifying mechanism is **NAD^+ deficiency during embryogenesis**, and crucially, malformations are **preventable by niacin (vitamin B3) supplementation** in animal models.

The disease is therefore best understood as a **gene × environment disorder with extreme variable expressivity**: identical biallelic KYNU genotypes can produce anything from an asymptomatic urinary metabolite trait to multisystem congenital malformations with developmental delay and premature death, depending substantially on maternal dietary NAD-precursor (tryptophan/niacin) status during pregnancy. This has profound and hopeful clinical implications — it is one of a small number of structural birth-defect syndromes that may be **preventable through maternal vitamin supplementation**.

Key Findings

Finding 1 — KYNU (kynureninase) deficiency causes xanthurenic aciduria / hydroxykynureninuria through a block in the tryptophan→NAD kynurenine pathway

The core molecular lesion of this disease is loss of kynureninase activity, which interrupts the kynurenine pathway and causes upstream metabolites to accumulate. The first molecular proof came from *Xanthurenic aciduria due to a mutation in KYNU encoding kynureninase* (PMID: 17334708), which reported that **"Massive urinary excretion of xanthurenic acid, 3-hydroxykynurenine and kynurenine, known as xanthurenic aciduria or hydroxykynureninuria, in a young Somali boy suggested kynureninase deficiency."** The index patient was homozygous for KYNU c.593A>G (p.T198A), and an affected younger brother carried the same genotype — establishing the gene–biochemistry link and the recessive inheritance.

This biochemical block was independently confirmed in *Biallelic variants in KYNU cause a multisystemic syndrome with hand hyperphalangism* (PMID: 31923704), in which **"Plasma and urine metabolomic analysis of two patients indicated a block along the tryptophan catabolic pathway and urine organic acid analysis showed excretion of xanthurenic acid."** The reported KYNU variants in that cohort included a large deletion of exons 1–8 combined with p.R428W, as well as p.R330Q and p.W109S — a mix of null and missense alleles.

Mechanistically, when kynureninase cannot cleave 3-hydroxykynurenine, 3-hydroxykynurenine accumulates and is diverted by transamination (via kynurenine aminotransferases) into **xanthurenic acid**, the hallmark urinary metabolite; kynurenine itself also accumulates and spills into urine. The accumulated/diverted metabolites define the diagnostic urine profile (CHEBI: xanthurenic acid CHEBI:10072; 3-hydroxy-L-kynurenine CHEBI:1547; L-kynurenine CHEBI:16946).

Finding 2 — Biallelic KYNU loss-of-function causes Congenital NAD Deficiency Disorder (CNDD/VCRL), rescuable by niacin in animal models

The landmark reconceptualization of this disease came from *NAD Deficiency, Congenital Malformations, and Niacin Supplementation* (PMID: 28792876, *New England Journal of Medicine*, Shi et al. 2017). In patients with combinations of cardiac, vertebral and renal defects, **"Variants were identified in two genes that encode enzymes of the kynurenine pathway, 3-hydroxyanthranilic acid 3,4-dioxygenase (HAAO) and kynureninase (KYNU)."** The specific KYNU alleles reported included p.V57Efs*21 and compound heterozygous p.Y156*/p.F349Kfs*4 — all loss-of-function. Mutant enzymes had greatly reduced in vitro activity, and patients had reduced circulating NAD.

The paper established the downstream mechanism explicitly: **"Nicotinamide adenine dinucleotide (NAD) is synthesized de novo from tryptophan through the kynurenine pathway. The patients had reduced levels of circulating NAD."** Critically, *Kynu*-null and *Haao*-null mouse embryos recapitulated the same constellation of malformations, which arose from gestational NAD deficiency — and **niacin supplementation during gestation prevented the malformations**. This simultaneously demonstrated causation (NAD deficiency is the proximate teratogenic driver) and identified a preventive intervention.

This finding redefined the disease: the "encephalopathy/hydroxykynureninuria" trait and the severe multisystem malformation syndrome are two ends of one NAD-deficiency spectrum arising from the same enzymatic block.

Finding 3 — CNDD is a gene × environment disorder with highly variable expressivity; prevention/treatment requires NAD-precursor (B3 vitamer) supplementation, with amidated precursors preferred for distal blocks

The gene × environment nature of this disorder — and the nuances of supplementation — are best demonstrated by work on the related pathway gene NADSYN1, which sits downstream and shares the CNDD phenotype. *A metabolic signature for NADSYN1-dependent congenital NAD*

deficiency disorder (PMID: 38357931, Szot et al. 2024) identified 13 further individuals with biallelic variants whose **"phenotypes ranging from multiple severe malformations to the complete absence of malformation"** — documenting the extraordinary variable expressivity that characterizes CNDD, even within families.

The same study established a therapeutically crucial principle about *which* B3 vitamer works, depending on where the pathway block lies: **"While for Nadsyn1+/- mothers, any B3 vitamer was suitable to raise NAD, preventing embryo loss and malformation, Nadsyn1-/- mothers required supplementation with amidated NAD precursors (nicotinamide or nicotinamide mononucleotide) bypassing their metabolic block."** By analogy, because KYNU acts upstream in the de novo (tryptophan) arm, an amidated precursor that enters via the salvage/Preiss-Handler route (nicotinamide, nicotinamide mononucleotide, nicotinamide riboside) bypasses the KYNU block and should be preferred over tryptophan or precursors requiring de novo synthesis.

Human evidence that supplementation raises the NAD pool comes from an adult NADSYN1-CNDD patient (PMID: 36649848) whose NAD pool rose approximately 25% after nicotinamide supplementation. The explicit framing of CNDD as a "gene × environment disorder caused by disruptions of the kynurenine pathway" appears in the 2026 review (PMID: 42798303).

Finding 4 — Kynureninase is a PLP-dependent fold-type I enzyme that strongly prefers 3-hydroxy-L-kynurenine over L-kynurenine

Understanding the enzyme's biochemistry explains the metabolite profile and informs therapeutic engineering. Human kynureninase (HsKYNase) is a **pyridoxal-5′-phosphate (PLP)-dependent** hydrolase of the aminotransferase (fold-type I) family. Protein-engineering studies note that the wild-type **"human enzyme (HsKYNase) has very low activity for kynurenine and shows no therapeutic effect"** (PMID: 36465553) — i.e., the physiological human enzyme preferentially cleaves **3-hydroxy-L-kynurenine** (to 3-hydroxyanthranilic acid + L-alanine) rather than L-kynurenine. This substrate preference explains why 3-hydroxykynurenine and its transamination product xanthurenic acid dominate the urinary signature in KYNU deficiency.

Catalysis depends on an intact PLP-binding pocket, and studies of engineered variants show that **"the amino acid substitutions in BF-HsKYNase allosterically affect the flexibility of the pyridoxal-5′-phosphate (PLP) binding pocket"** (PMID: 35658075), modulating the rate-limiting chemical step. This has two implications for the disease: (1) missense variants that perturb PLP binding or active-site geometry are plausibly pathogenic through catalytic loss of

function; and (2) the enzyme's PLP dependence connects the pathway to vitamin B6 status (pyridoxine), historically relevant to xanthurenic acid excretion (xanthurenic aciduria is also a classic marker of B6 deficiency).

Finding 5 — KYNU deficiency presents on a spectrum from asymptomatic xanthurenic aciduria to VCRL malformations with hand hyperphalangism, mimicking VACTERL / Catel-Manzke syndrome

The clinical phenotype is remarkably broad. At the mild end, *Xanthurenic aciduria due to a mutation in KYNU* (PMID: 17334708) described two brothers with the full biochemical phenotype who had **"neither... symptoms of niacin deficiency"** — essentially an asymptomatic metabolic trait. At the severe end, *Biallelic variants in KYNU cause a multisystemic syndrome with hand hyperphalangism* (PMID: 31923704) described **"three patients with hand hyperphalangism, heart defect, short stature, and mild to severe developmental delay, all of whom were initially given a clinical diagnosis of Catel-Manzke syndrome."** Notably, a multi-exon KYNU deletion (exons 1–8) was detectable only with copy-number variant analysis — a diagnostic lesson.

The overlap with recognized malformation associations is substantial. *Further description of two patients with biallelic variants in NADSYN1* (PMID: 35491967, Kortbawi et al. 2022) notes that NAD-deficiency disorders **"overlap with the anomalies present in vertebral, anal, cardiac, tracheoesophageal, radial and renal, and limb anomalies (VATER/VACTERL) association and often result in premature death; children who survive typically have developmental delays or intellectual disability."** Thus KYNU deficiency can masquerade as VACTERL association or Catel-Manzke syndrome, and should be considered in those differential diagnoses — especially because, unlike most causes of those phenotypes, it is biochemically detectable and potentially preventable.

Mechanistic Model / Interpretation

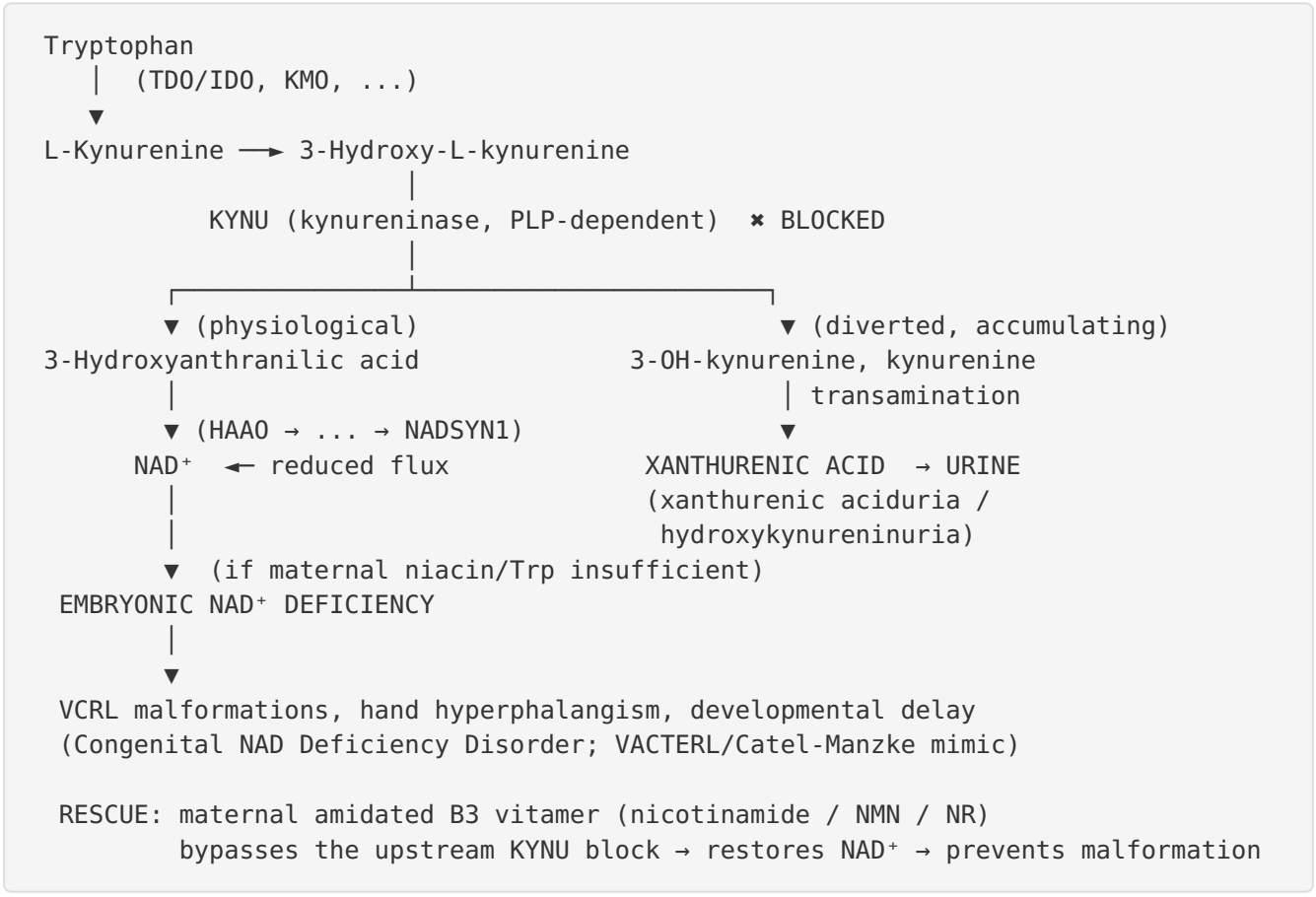
Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function variants in KYNU** (nonsense, frameshift, large deletion, or damaging missense affecting the PLP pocket) **lead to** absent or severely reduced kynureninase enzyme activity.

2. Loss of kynureninase activity **results in** a block in the de novo kynurenine pathway at the hydrolysis of 3-hydroxy-L-kynurenine → 3-hydroxyanthranilic acid (the physiologically preferred reaction of the human enzyme).
3. The block **causes** accumulation of upstream metabolites — 3-hydroxykynurenine and kynurenine — which **branch** into two consequences:
4. **Branch A (biochemical/diagnostic):** accumulated 3-hydroxykynurenine is transaminated to **xanthurenic acid**, and kynurenine/3-hydroxykynurenine spill into urine → **xanthurenic aciduria** / **hydroxykynureninuria** (the measurable trait).
5. **Branch B (pathogenic):** reduced downstream flux **leads to** decreased de novo synthesis of **NAD^+**.
6. During embryogenesis, if maternal dietary NAD precursors (tryptophan, niacin/nicotinamide) are insufficient to compensate, the NAD^+ shortfall **results in** a critical deficiency of NAD^+ in the developing embryo (inferred to impair NAD^+-dependent processes — energy metabolism, PARP/sirtuin signaling, redox balance — during organogenesis; the precise molecular targets in human embryos remain inferred rather than fully demonstrated).
7. Embryonic NAD^+ deficiency during organogenesis **causes** congenital malformations — **vertebral, cardiac, renal and limb (VCRL)** defects, hand hyperphalangism, and in severe cases miscarriage/premature death — i.e., **Congenital NAD Deficiency Disorder (CNDD)**.
8. Postnatally, persistent NAD-pathway dysfunction and/or the malformations themselves **contribute to** developmental delay and intellectual disability in survivors.

Gene × environment modifier (acts on steps 3B–5): Maternal NAD-precursor status is the dominant environmental modifier. Abundant maternal niacin/tryptophan **can prevent** the NAD^+ shortfall and therefore the malformations, even with the same biallelic genotype — explaining the spectrum from asymptomatic xanthurenic aciduria to severe multisystem malformation.

Schematic



Upstream vs downstream

- **Upstream (initiating):** KYNU genotype → loss of kynureninase catalysis → metabolite block.
- **Midstream (dual output):** metabolite accumulation (→ diagnostic xanthurenic aciduria) AND reduced NAD^+ synthesis.
- **Downstream (clinical):** embryonic NAD^+ deficiency → malformations; postnatal developmental delay.
- **Dominant modifier:** maternal/dietary NAD-precursor availability determines whether the downstream clinical branch is expressed.

Ontology suggestions

- **Gene/protein:** KYNU (HGNC:6469); GO biological process: *NAD biosynthetic process* (GO:0009435), *tryptophan catabolic process to kynurenine* (GO:0019441), *quinolinate biosynthetic process* (GO:0019805); GO molecular function: *kynureninase activity*

(GO:0030429), *pyridoxal phosphate binding* (GO:0030170); GO cellular component: *cytosol* (GO:0005829).

- **Phenotypes (HPO):** Vertebral segmentation defect (HP:0003422), Abnormal heart morphology / congenital heart defect (HP:0001627), Renal hypoplasia/agenesis (HP:0008678), Short stature (HP:0004322), Global developmental delay (HP:0001263), Intellectual disability (HP:0001249), Hyperphalangy (HP:0009811), Aminoaciduria (HP:0003355) as nearest general term for the urinary metabolite abnormality.
- **Anatomy (UBERON):** vertebral column (UBERON:0001130), heart (UBERON:0000948), kidney (UBERON:0002113), limb (UBERON:0002101), brain (UBERON:0000955).
- **Chemical (CHEBI):** xanthurenic acid (CHEBI:10072), 3-hydroxy-L-kynurenine (CHEBI:1547), L-kynurenine (CHEBI:16946), NAD^+ (CHEBI:15846), nicotinamide (CHEBI:17154), niacin/nicotinic acid (CHEBI:15940), pyridoxal 5′-phosphate (CHEBI:18405).
- **Treatment (NCIT):** Nicotinamide (NCIT:C577), Niacin/Nicotinic Acid (NCIT:C545), Dietary supplement / Nutritional support.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports
17334708	Xanthurenic aciduria due to a mutation in KYNU	Human clinical/ genetic	First molecular proof: KYNU mutation → xanthurenic aciduria; asymptomatic trait end of spectrum (F1, F5)
28792876	NAD Deficiency, Congenital Malformations, Niacin Supplementation (NEJM)	Human + mouse	KYNU/HAAO LoF → reduced NAD → VCRL malformations; niacin prevents in mice (F2)
31923704	Biallelic KYNU variants cause multisystemic syndrome with hand hyperphalangism	Human clinical/ genetic/ metabolomic	Confirms tryptophan-pathway block; Catel-Manzke mimic; CNV detection (F1, F5)
38357931	Metabolic signature for NADSYN1-dependent CNDD	Human + mouse	Variable expressivity; amidated precursors required for distal block (F3)
36649848	Adult NADSYN1-associated CNDD, NAD analysis	Human clinical	Nicotinamide supplementation raised NAD pool ~25% (F3)
42798303	Biallelic variants … CNDD (2026 review)	Review	CNDD framed as gene × environment disorder of kynurenine pathway (F3)
36465553	Switching the rate-limiting step of an immunotherapeutic enzyme	In vitro / protein engineering	Human KYNase prefers 3-OH-kynurenine; low activity for kynurenine (F4)
35658075	Leveraging intrinsic flexibility to engineer enzyme activity	In vitro / structural	KYNU is PLP-dependent; PLP-pocket flexibility controls catalysis (F4)
35491967	Two patients with biallelic NADSYN1 variants	Human clinical	VACTERL overlap; premature death; developmental delay in survivors (F5)
35484986	NAD^+ deficiency in human malformations and miscarriage: pleiotropy model	Review/ hypothesis	Broad CNDD pleiotropy framework (context)
39829932	Zebrafish model of NAD deficiency disorder	Model organism	Additional non-mammalian CNDD model (context)

PMID	Title (abbrev.)	Evidence type	Supports
37300479	NAD Deficiency and Mammalian Development	Review	Developmental role of NAD (context)

Historical/primary descriptions (pre-molecular era): *Hydroxykynureninuria* (PMID: 6015911); *Hydroxykynureninuria: abnormal tryptophan metabolism probably due to kynureninase deficiency* (PMID: 14169454); *Familial hydroxykynureninuria* (PMID: 649181); *Hydroxykynureninuria, a case report* (PMID: 7152601). These 1960s–1980s reports established the biochemical phenotype and inferred kynureninase deficiency decades before KYNU was sequenced.

Section-by-Section Compilation

1. Disease Information

Encephalopathy due to hydroxykynureninuria (xanthurenic aciduria; kynureninase deficiency) is a rare autosomal-recessive inborn error of tryptophan catabolism. **Identifiers:** MONDO:0009372; OMIM 236800; gene KYNU (HGNC:6469); Orphanet lists it within kynurenine-pathway/CNDD disorders; ICD-10 maps best to E70.8 (other disorders of aromatic amino-acid metabolism). The modern literature increasingly subsumes it under **Congenital NAD Deficiency Disorder (CNDD)**. Information is derived from a combination of **aggregated disease-level resources** (OMIM, Orphanet) and **individual patient reports/case series** (the KYNU and NADSYN1/HAAO cohorts). The name "encephalopathy" is partly historical; the predominant modern phenotype is congenital malformation ± neurodevelopmental delay rather than a classic progressive encephalopathy.

2. Etiology

Causal factor: biallelic (homozygous or compound heterozygous) loss-of-function variants in **KYNU**. **Genetic risk factors:** the only established determinant is biallelic pathogenic KYNU genotype; **consanguinity** increases risk (several reported families are consanguineous; e.g., the homozygous Somali brothers). **Environmental risk / protective factors:** maternal dietary **NAD-precursor (tryptophan, niacin/nicotinamide) deficiency** during pregnancy is the major environmental factor that converts a biochemical trait into congenital malformations;

conversely, **adequate maternal niacin is protective** (demonstrated in mouse models). **Gene–environment interaction** is the defining feature: genotype sets susceptibility, maternal NAD-precursor status determines expression.

3. Phenotypes

- **Laboratory abnormalities (obligate):** xanthurenic aciduria, elevated urinary 3-hydroxykynurenine and kynurenine, reduced circulating NAD (nearest HPO: Aminoaciduria HP:0003355).
- **Structural (variable):** vertebral segmentation defects (HP:0003422), congenital heart defects (HP:0001627), renal hypoplasia/agenesis (HP:0008678), limb/hand anomalies including hyperphalangy (HP:0009811), short stature (HP:0004322).
- **Neurodevelopmental:** global developmental delay (HP:0001263), intellectual disability (HP:0001249) in survivors of the severe form.
- **Onset:** congenital/prenatal for malformations; the biochemical trait is present from birth. **Severity:** highly variable (asymptomatic → severe/lethal). **Frequency among affected:** metabolite abnormalities ~obligate; malformations present in the severe CNDD subset. **QoL:** severe end entails major disability and premature death; mild end may be clinically silent.

4. Genetic / Molecular Information

Causal gene: KYNU (OMIM *605197; HGNC:6469; chr 2q22.2). Reported pathogenic variants include missense (p.T198A, p.R428W, p.R330Q, p.W109S), nonsense (p.Y156*), frameshift (p.V57Efs*21, p.F349Kfs*4), and a **large multi-exon deletion (exons 1–8)** detectable only by CNV analysis. **Variant classification:** pathogenic/likely pathogenic per ACMG; most are loss-of-function. **Allele frequency:** individually very rare in gnomAD (consistent with a rare recessive disorder). **Origin:** germline. **Functional consequence:** loss of function (reduced/absent kynureninase catalytic activity). **Modifier genes:** other kynurenine-pathway genes (HAAO, NADSYN1) cause the same CNDD phenotype and may act as pathway-level modifiers; no specific intragenic modifier established. **Epigenetic/chromosomal abnormalities:** none specifically implicated beyond the structural KYNU deletion.

5. Environmental Information

The key environmental factor is **maternal dietary NAD-precursor status** (tryptophan and niacin/vitamin B3). Low dietary niacin unmasks malformations; sufficient intake is protective. Vitamin **B6 (pyridoxine)** status is mechanistically relevant because kynureninase is PLP-dependent, and xanthurenic aciduria is also a classic marker of B6 deficiency. No infectious agents are involved.

6. Mechanism / Pathophysiology

See the **Mechanistic Model** section above for the full numbered causal chain. In brief: KYNU LoF → kynureninase block → (a) accumulation/diversion of 3-hydroxykynurenine to xanthurenic acid (diagnostic) and (b) reduced de novo NAD^+ synthesis → embryonic NAD^+ deficiency (gated by maternal precursor supply) → VCRL malformations and developmental delay. **Pathways:** kynurenine/NAD de novo pathway (KEGG hsa00380 tryptophan metabolism; Reactome NAD biosynthesis). **Biochemical defect:** PLP-dependent kynureninase deficiency. **Metabolic change:** impaired amino-acid (tryptophan) catabolism and NAD^+ cofactor depletion.

7. Anatomical Structures Affected

Primary organs/systems: axial skeleton/vertebral column (UBERON:0001130), heart (UBERON:0000948), kidneys/renal system (UBERON:0002113), limbs (UBERON:0002101); nervous system (UBERON:0000955) for neurodevelopmental involvement. **Lateralization:** malformations are typically bilateral/midline (vertebral, cardiac, renal). **Subcellular:** cytosolic enzyme (kynureninase, GO:0005829); NAD^+ depletion affects mitochondrial and nuclear NAD-dependent processes. **Cell/tissue level:** broadly affects dividing embryonic progenitor populations during organogenesis (no single CL cell type is specifically targeted; the effect is a systemic cofactor deficiency).

8. Temporal Development

Onset: congenital (malformations arise during embryonic organogenesis; biochemical trait present from birth). **Pattern:** the malformation damage is fixed at birth (non-progressive structurally), but neurodevelopmental sequelae and growth deficits manifest over childhood. **Critical period:** the therapeutic window is **periconceptional/gestational** — maternal supplementation must occur during pregnancy to prevent malformations; postnatal supplementation cannot reverse structural defects. **Course:** severe cases may be lethal prenatally/neonatally; mild cases are lifelong but asymptomatic.

9. Inheritance and Population

Inheritance: autosomal recessive. **Penetrance/expressivity:** highly variable expressivity (asymptomatic trait → severe malformation/miscarriage), modified by maternal environment; biochemical penetrance is high. **Consanguinity:** contributory (reported consanguineous and founder-type families). **Epidemiology:** ultra-rare; precise prevalence/incidence not established (only a few dozen molecularly confirmed CNDD cases across KYNU/HAAO/NADSYN1). **Sex ratio:** no established predilection. **Carrier frequency:** not precisely defined; individual alleles are rare in gnomAD.

10. Diagnostics

Biochemical: urine organic acid / metabolomic analysis showing xanthurenic acid, 3-hydroxykynurenine and kynurenine excretion; plasma/urine metabolomics showing a tryptophan-pathway block; reduced circulating NAD. **Genetic testing:** single-gene KYNU testing, gene panels covering kynurenine/NAD-pathway genes (KYNU, HAAO, NADSYN1), or WES/WGS — with the important caveat that **CNV/deletion analysis is required** because large KYNU deletions (exons 1–8) are missed by SNV-only sequencing. **Imaging:** skeletal survey (vertebral segmentation), echocardiography (cardiac defects), renal ultrasound. **Differential diagnosis:** VACTERL/VATER association, Catel-Manzke syndrome, and other causes of vertebral/cardiac/renal/limb malformations — distinguished by the characteristic urinary metabolite signature and KYNU genotype.

11. Outcome / Prognosis

Prognosis spans the full range: asymptomatic biochemical trait with normal life expectancy at the mild end; at the severe end, CNDD "often results in premature death," and survivors typically have developmental delay/intellectual disability (PMID: 35491967). Prognosis is largely determined prenatally; the prognostic modifier is the severity/number of organ malformations. No formal survival statistics exist given rarity.

12. Treatment

Primary/preventive: maternal **NAD-precursor (vitamin B3) supplementation** during pregnancy to prevent malformations — the central therapeutic insight. Because KYNU acts upstream, **amidated precursors (nicotinamide, nicotinamide mononucleotide, nicotinamide riboside)** that bypass the de novo block are preferred over tryptophan (by analogy to the NADSYN1 data, PMID: 38357931). Human data show nicotinamide raises the NAD pool (~25%, PMID: 36649848). **Postnatal:** supportive and surgical management of

malformations (cardiac surgery, orthopedic/vertebral and renal management), developmental/rehabilitative support. No approved gene therapy or targeted drug. **NCIT terms:** Nicotinamide (NCIT:C577), Niacin (NCIT:C545). **PLP/vitamin B6** repletion is a reasonable adjunct given the PLP dependence of the enzyme, though it cannot restore a null enzyme.

13. Prevention

Primary prevention: periconceptional/gestational maternal niacin/nicotinamide supplementation in at-risk pregnancies (standard prenatal vitamin niacin content may be insufficient for biallelic carriers; higher-dose amidated precursors are rational). **Secondary:** carrier screening in consanguineous families; prenatal/preimplantation genetic testing where the familial variants are known; biochemical screening (urinary xanthurenic acid). **Genetic counseling:** autosomal-recessive recurrence risk (25%), with counseling on the potentially preventable nature of malformations through maternal supplementation.

14. Other Species / Natural Disease

Model/ortholog species: mouse (*Kynu*; NCBI Gene) and zebrafish NAD-pathway models recapitulate CNDD. Kynureninase is evolutionarily conserved; bacterial kynureninases (e.g., in oral and mosquito gut microbiota) perform the same 3-hydroxykynurenine → 3-hydroxyanthranilic acid reaction (PMID: 35437328, PMID: 40988889), underscoring deep conservation of the enzymatic step. No naturally occurring companion-animal disease is specifically documented for KYNU deficiency.

15. Model Organisms

- **Mouse:** *Kynu*-null and *Haao*-null embryos develop VCRL-type malformations from NAD deficiency, and **niacin supplementation during gestation prevents them** (PMID: 28792876) — the strongest causal/preventive model. *Nadsyn1* mouse models further dissect precursor requirements (PMID: 38357931).
- **Zebrafish:** a CNDD model recapitulates multisystem anomalies (PMID: 39829932).
- **Recapitulation:** models reproduce the malformation spectrum and the niacin-rescue phenotype well; **limitation:** they less directly model the human neurodevelopmental/"encephalopathy" component and the full range of mild human presentations.

Limitations and Knowledge Gaps

1. **Few molecularly confirmed KYNU patients.** The evidence base rests on a handful of families; precise prevalence, penetrance, sex ratio, and genotype–phenotype correlations are not statistically defined.
2. **"Encephalopathy" nomenclature vs. reality.** The historical name implies a progressive encephalopathy, but the modern molecular phenotype is congenital malformation ± neurodevelopmental delay. The relationship between the original 1960s "hydroxykynureninuria" cases and the severe CNDD phenotype is a spectrum unified by mechanism, but the neurological manifestations remain under-characterized.
3. **Human embryonic mechanism inferred.** The step from NAD^+ deficiency to specific malformations is demonstrated in mice/zebrafish; the exact NAD^+-dependent molecular targets in human embryogenesis are inferred, not directly proven.
4. **Supplementation regimen for KYNU specifically.** Optimal precursor, dose, and timing for human KYNU-deficient pregnancies are extrapolated from NADSYN1/mouse data; no KYNU-specific clinical trial exists.
5. **Modifier genetics.** Whether specific modifier alleles (beyond maternal diet) tune expressivity is unknown.

Proposed Follow-up Experiments / Actions

1. **Registry and natural history study** of all molecularly confirmed KYNU/HAAO/NADSYN1 (CNDD) patients to quantify penetrance, phenotype frequencies, and long-term neurodevelopmental outcomes.
2. **Prospective maternal NAD-metabolome profiling** in at-risk pregnancies (known biallelic KYNU carriers) to correlate maternal NAD/precursor levels with offspring outcome and define a protective threshold.
3. **Controlled supplementation trial** (nicotinamide vs. nicotinamide riboside, dose-finding) in mouse *Kynu*-null dams to define the KYNU-specific optimal amidated-precursor regimen and critical gestational window.
4. **Functional characterization** of each reported KYNU missense variant (p.T198A, p.R428W, p.R330Q, p.W109S) via recombinant enzyme kinetics and PLP-binding assays, leveraging the structural insights from PMID: 35658075 and PMID: 36465553, to build a genotype–activity–phenotype map.

5. **Diagnostic guideline:** promote urinary xanthurenic-acid metabolomic screening + KYNU CNV analysis in the VACTERL/Catel-Manzke diagnostic workup, given the preventable nature of the disorder and the risk of missing large deletions.
6. **Preconception counseling protocol** for consanguineous families with a prior affected child, incorporating high-dose amidated B3 supplementation during subsequent pregnancies.

Evidence-source legend: Human clinical/genetic (17334708, 31923704, 28792876, 35491967, 36649848); mouse model (28792876, 38357931); zebrafish model (39829932); in vitro/structural (36465553, 35658075); review/hypothesis (42798303, 35484986, 37300479).
